

STRUCTURAL ORGANISATION IN ANIMALS

Section 7.1: Organ & Organ System • Section 7.2: Comprehensive Morphology & Anatomy of Frog (*Rana tigrina*)

CHAPTER MIND MAP — ARCHITECTURE OF CHAPTER 7

[Hierarchy Overview](#)

7.1 Organ & System

Cells → Tissues → Organs → Organ Systems • Division of Labour • Evolutionary Trend

7.2 Frog Overview

Rana tigrina • Poikilotherm • Camouflage / Mimicry • Aestivation & Hibernation

7.2.1 Morphology

Head & Trunk • Moist Skin • Bulged Eyes & Nictitating Membrane • Sexual Dimorphism

7.2.2 Organ Systems

Digestive, Respiratory, Vascular, Excretory, Neural, Endocrine & Sense Organs

Reproduction & Role

Male & Female Systems • Bidder's Canal • Metamorphosis • Ecological Importance

7.1 ORGAN AND ORGAN SYSTEM

In unicellular organisms, all vital functions (digestion, respiration, reproduction) are performed by a single cell. In multicellular animals, these basic functions are carried out by distinct groups of organized cells.

Definition of Tissue

"In multicellular animals, a group of similar cells along with intercellular substances performing a specific function is called a **tissue**."

Cellular Abundance:

- Hydra*: Made of thousands of cells of different types.
- Human Body*: Composed of **billions of cells** working together.

4 Basic Tissue Types:

All complex animals consist of only **four basic tissue types** organized in specific proportions and patterns to form organs (e.g., stomach, lungs, heart, kidneys).

Biological Hierarchy in Multicellular Organisms

- 1. Cells**: Basic structural & functional units.
- 2. Tissues**: Similar cells + intercellular matrix.
- 3. Organs**: Specific proportion of 4 tissue types.
- 4. Organ Systems**: Organs working interactively.
- 5. Organism**: Integrated survival machine.

Easy Way to Remember: 4 Heart Tissues

Memory Trick: "Every Child Must Nutritate"

E - Epithelial Tissue
C - Connective Tissue
M - Muscular Tissue
N - Neural Tissue

Division of Labour & Evolutionary Trend

When two or more organs perform a common function through physical and/or chemical interactions, they form an **organ system** (e.g., digestive, respiratory). Cells, tissues, organs, and organ systems exhibit **division of labour** to contribute to body survival.

★ **Heart Composition Example:** The human heart consists of **all four basic tissue types**: Epithelial, Connective, Muscular, and Neural tissues. Organ complexity displays a clear **evolutionary trend**.

🌳 Taxonomic Classification & Species

• Freshwater & Land (Amphibious)

Phylum	Chordata
Class	Amphibia
Common Species (India)	<i>Rana tigrina</i>
Habitat	

Definition of Amphibian: Can live both on land and in freshwater. Represents class Amphibia of phylum Chordata.

🦎 Key Physiological & Defensive Adaptations

1. Poikilotherms

Cold-blooded animals: Do not have a constant body temperature; body temp varies with the environment.

2. Camouflage & Mimicry

Ability to change skin color to hide from enemies in grass/dry land. Protective coloration is called **mimicry**.

3. Dormancy Period

Take shelter in deep burrows during peak summer (**Aestivation**) and peak winter (**Hibernation**).

🏠 7.2.1 EXTERNAL MORPHOLOGY OF FROG

EXTERNAL ANATOMY

Skin & Coloration Characteristics

- **Texture:** Smooth and slippery due to mucus secretion.
- **Moisture:** Skin is always maintained in a moist condition.
- **Water Intake:** Frog **never drinks water** directly; absorbs water through the skin.
- **Dorsal Side Color:** Olive green with dark irregular spots.
- **Ventral Side Color:** Uniformly pale yellow.

Body Divisions & Sensory Features

- **Divisions:** Divisible into **Head** and **Trunk**.
- **Absent Parts:** **Neck** and **Tail** are absent.
- **Nostrils:** A pair of nostrils present above the mouth.
- **Eyes:** Bulged, covered by a **nictitating membrane** protecting eyes in water.
- **Ears:** Membranous **tympanum** on either side receives sound signals.

Limbs & Adaptations

- **Functions:** Swimming, walking, leaping, and burrowing.
- **Forelimbs:** End in **4 digits**; smaller than hindlimbs.
- **Hindlimbs:** End in **5 digits**; larger and more muscular.
- **Webbed Feet:** Digits of feet are webbed to aid in swimming.

♂️ SEXUAL DIMORPHISM IN FROGS

HIGH YIELD EXAM TABLE

Morphological Feature	Male Frog	Female Frog
Vocal Sacs	PRESENT (Produces croaking sound for mating)	ABSENT

🍴 DIGESTIVE SYSTEM & PROCESS

CARNIVOROUS ADAPTATION

Short Alimentary Canal:

The alimentary canal of frog is **short** because frogs are **carnivores** and hence the length of the intestine is reduced.

Path of Food Passage:

Mouth → Buccal Cavity

Pharynx →
Oesophagus

Stomach → Duodenum

Intestine → Rectum →
Cloaca

Organ / Gland	Structure & Secretion	Specific Digestive Function
Tongue	Bilobed tongue	Captures prey efficiently.
Stomach	Secretes HCl & Gastric juices	Digests food into partially digested liquid mixture called Chyme .
Liver	Secretes Bile (stored in Gall Bladder)	Emulsifies fats in the duodenum.
Pancreas	Produces Pancreatic juice	Enzymes digest carbohydrates and proteins .
Intestine	Inner wall has Villi & Microvilli	Final digestion & absorption of digested food. Undigested waste goes to rectum & exits via cloaca .

🌬️ RESPIRATORY SYSTEM

DUAL RESPIRATION

Frogs exhibit two distinct modes of respiration based on habitat (water vs land).

1. Cutaneous Respiration (In Water)

Skin

Dissolved oxygen in water is exchanged through skin by simple **diffusion**. Skin is kept moist & highly vascularised.

2. Pulmonary & Buccal Respiration (On Land)

Lungs / Mouth

On land, buccal cavity, skin, and lungs act as respiratory organs. Lungs are a pair of **elongated, pink-coloured sac-like structures** in upper trunk (thorax). Air enters: Nostrils → Buccal Cavity → Lungs.

⚠️ Respiration During Dormancy

During **aestivation** (summer sleep) and **hibernation** (winter sleep), gaseous exchange takes place **EXCLUSIVELY** through the **skin**.

★ ★ High-Yield Exam Points: Chyme, Common Bile Duct & Respiratory Organ Summary

Chyme: Partially digested, acidic food passed from stomach to duodenum.

Common Bile Duct: Conveys bile and pancreatic juice into duodenum together.

Dormancy Respiration: Lungs & Buccal cavity inactive; 100% skin diffusion.

♥ BLOOD VASCULAR & LYMPHATIC SYSTEM

CLOSED TYPE

Heart Anatomy:

- **3-Chambered:** 2 Atria + 1 Ventricle.
- Covered by membrane called **Pericardium**.
- **Sinus Venosus:** Triangular structure joining right atrium; receives blood from vena cava.
- **Conus Arteriosus:** Sac-like structure on ventral side where ventricle opens.

Portal Systems:

- **Hepatic Portal System:** Special venous connection between **liver and intestine**.
- **Renal Portal System:** Special venous connection between **kidney and lower body parts**.

Blood Component	Structural Features & Characteristics
Erythrocytes (RBCs)	Nucleated and contain red pigment Haemoglobin .
Leucocytes (WBCs) & Platelets	Present in blood plasma; defensive and clotting functions.
Lymph	Different from blood; lacks RBCs and few proteins .

▼ EXCRETORY SYSTEM

UREOTELIC ANIMAL

Excretory Organs:

Consists of a pair of **kidneys**, **ureters**, **cloaca**, and a **urinary bladder**.

- **Kidney Appearance:** Compact, dark red, bean-like structures situated posteriorly on both sides of vertebral column.
- **Functional Unit:** Composed of uriniferous tubules or **nephrons**.
- **Excretory Product:** Excretes **Urea** (Ureotelic). Wastes carried by blood to kidneys.
- **Urinary Bladder:** Thin-walled, ventral to rectum, opens into **cloaca**.

🔗 Crucial Sex Difference in Ureters:

Male Frog: Ureters act as **Urinogenital duct** (carries urine + sperms) opening into cloaca.

Female Frog: Ureters & oviducts open **separately** into cloaca.

💡 🗨 Easy Way to Remember: Portal Systems & RBC Nucleation

- **Frog RBCs have Nuclei** (Unlike mature Human RBCs which are enucleated!).
- **Portal Systems:** Hepatic = Liver-Intestine | Renal = Kidney-Lower Body. Both enter organ before going to heart.

ENDOCRINE GLANDS LIST**HORMONES**

Chemical coordination is achieved by hormones secreted by prominent endocrine glands:

- | | |
|----------------|----------------------|
| 1. Pituitary | 2. Thyroid |
| 3. Parathyroid | 4. Thymus |
| 5. Pineal Body | 6. Pancreatic Islets |
| 7. Adrenals | 8. Gonads |

NERVOUS SYSTEM & BRAIN REGIONS**10 PAIRS CRANIAL NERVES****CNS:** Brain & Spinal Cord**PNS:** Cranial & Spinal Nerves (**10 Pairs Cranial Nerves**)**ANS:** Sympathetic & Parasympathetic

Brain Division	Constituent Structures & Anatomic Features
Fore-brain	Includes Olfactory lobes (smell), paired Cerebral hemispheres , and unpaired Diencephalon .
Mid-brain	Characterised by a pair of prominent Optic lobes (vision).
Hind-brain	Consists of Cerebellum and Medulla oblongata . Medulla passes out through Foramen Magnum into spinal cord (in vertebral column). Enclosed in bony Cranium (Brain box) .

5 SENSE ORGANS OF FROG & ORGANISATION**SIMPLE EYES & EARS**

- Tactile receptors located in skin.
- Located in buccal cavity.
- Olfactory receptors in nasal chamber.

Sense Modality	Sense Organ	Structural Complexity	Key Features & Functional Notes
Touch	Sensory Papillae	Cellular aggregations around nerve endings	
Taste	Taste Buds	Cellular aggregations around nerve endings	
Smell	Nasal Epithelium	Cellular aggregations around nerve endings	
Vision	Eyes	Well-Organised Structure	Pair of spherical structures in orbit of skull. Simple eyes (possessing only one unit).
Hearing & Balance	Tympanum + Internal Ear	Well-Organised Structure	External ear absent. Ear serves dual function: Hearing AND Balancing (Equilibrium) .

♂ MALE REPRODUCTIVE SYSTEM

BIDDER'S CANAL

- **Testes:** Pair of yellowish ovoid testes attached to upper part of kidneys by double fold of peritoneum called **Mesorchium**.
- **Vasa Efferentia:** 10–12 in number, arise from testes, enter kidneys and open into **Bidder's canal**.
- **Urinogenital Duct:** Communicates with Bidder's canal, emerges from kidney to open into **Cloaca**.
- **Cloaca Function:** Small median chamber passing faecal matter, urine, and sperms to exterior.

Sperm Passage Route:

Testes → Vasa Efferentia (10-12) → Kidney → Bidder's Canal → Urinogenital Duct → Cloaca → Exterior.

♀ FEMALE REPRODUCTIVE SYSTEM

2500–3000 OVA

- **Ovaries:** Pair of ovaries situated near kidneys. **NO functional connection with kidneys**.
- **Oviducts:** Pair of oviducts arising from ovaries open **separately** into the cloaca.
- **Egg Laying Capacity:** A mature female lays **2500 to 3000 ova** at a time.
- **Fertilisation:** **External**, takes place in water.
- **Development:** Indirect through larval stage called **Tadpole** which undergoes **metamorphosis** into adult.

Key Distinguishing Fact:

Unlike males, female ovaries have NO connection with kidneys. Ureters & Oviducts open independently into cloaca.

🌿 ECOLOGICAL & ECONOMIC IMPORTANCE OF FROGS

1. Crop Protection: Eat insects and protect agricultural crops from pests.

2. Ecosystem Balance: Important link in food chain and food web maintaining ecological balance.

3. Human Food Source: Muscular legs used as food by humans in certain countries.

🔄 📄 FINAL QUICK REVISION — RAPID RECALL

EXAM CHECKLIST

Heart: 3 Chambers (2A + 1V), Sinus Venosus (Rt Atrium), Conus Arteriosus.

Nerves: 10 pairs cranial nerves; Cranium protects brain.

RBCs: Nucleated + Haemoglobin; Lymph lacks RBCs & few proteins.

Excretion: Ureotelic (Urea); Nephrons in bean-like kidneys.

